

CHAPTER – 4

FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES ANALYSIS

4.1 INTRODUCTION

Functional Dependency (FD) is a relationship between attributes where one attribute uniquely determines another. Analyzing functional dependencies helps identify redundancy and prevents update, insertion, and deletion anomalies. The following analysis explains the functional dependencies, redundancy, and possible anomalies for each entity in the Purple Company E-Commerce Database System.

4.2 CUSTOMER

Attribute	Description
Customer_ID (PK)	Unique customer ID
Customer_Name	Customer name
Email	Email address
Phone_Number	Mobile number
Address	Customer delivery address

Functional Dependency

- Customer_ID → Customer_Name, Email, Phone_Number, Address

Redundancy

Customer information may be duplicated if the same customer is registered multiple times.

Data Anomalies

No data anomaly is present because customer details are uniquely stored using Customer_ID.

4.3 CATEGORY

Attribute	Description
Category_ID (PK)	Unique category ID
Category_Name	Product category name

Functional Dependency

- Category_ID → Category_Name

Redundancy

Category details may be repeated for multiple products if the database is not properly normalized.

Data Anomalies

No data anomaly is present because category details are uniquely stored using Category_ID.

4.4 PRODUCT

Attribute	Description
Product_ID (PK)	Unique product ID
Product_Name	Product name
Brand	Brand name
Price	Product price
Stock	Available stock
Category_ID (FK)	Product category

Functional Dependency

- Product_ID → Product_Name, Brand, Price, Stock, Category_ID

Redundancy

Product information may be duplicated if the same product is entered multiple times.

Data Anomalies

No data anomaly is present because each product is uniquely stored using Product_ID.

4.5 ORDERS

Attribute	Description
Order_ID (PK)	Unique order ID
Customer_ID (FK)	Customer ID
Order_Date	Order date
Total_Amount	Total amount
Order_Status	Current order status

Functional Dependency

- Order_ID → Customer_ID, Order_Date, Total_Amount, Order_Status

Redundancy

Order details may be duplicated if the same order is recorded multiple times.

Data Anomalies

No data anomaly is present because each order is uniquely stored using Order_ID.

4.6 ORDER_ITEM

Attribute	Description
Order_Item_ID (PK)	Unique order item ID
Order_ID (FK)	Order ID
Product_ID (FK)	Product ID
Quantity	Quantity ordered
Price	Product price

Functional Dependency

- Order_Item_ID → Order_ID, Product_ID, Quantity, Price

Redundancy

Order item details may be duplicated if the same order item is entered multiple times.

Data Anomalies

No data anomaly is present because each order item is uniquely stored using Order_Item_ID.

4.7 PAYMENT

Attribute	Description
Payment_ID (PK)	Unique payment ID
Order_ID (FK)	Order ID
Payment_Method	Payment method
Payment_Status	Payment status

Functional Dependency

- Payment_ID → Order_ID, Payment_Method, Payment_Status

Redundancy

Payment information may be duplicated if identical payment records are entered multiple times.

Data Anomalies

No data anomaly is present because each payment is uniquely stored using Payment_ID.

4.8 DELIVERY

Attribute	Description
Delivery_ID (PK)	Unique delivery ID
Order_ID (FK)	Order ID
Tracking_Number	Shipment tracking number
Delivery_Status	Delivery status

Functional Dependency

- Delivery_ID → Order_ID, Tracking_Number, Delivery_Status

Redundancy

Delivery details may be duplicated if shipment information is entered multiple times.

Data Anomalies

No data anomaly is present because each delivery record is uniquely stored using Delivery_ID.

4.9 SUPPLIER

Attribute	Description
Supplier_ID (PK)	Unique supplier ID
Supplier_Name	Supplier name
Contact_Number	Contact number
Email	Supplier email

Functional Dependency

- Supplier_ID → Supplier_Name, Contact_Number, Email

Redundancy

Supplier information may be duplicated if the same supplier is entered multiple times.

Data Anomalies

No data anomaly is present because supplier details are uniquely stored using Supplier_ID.

4.10 ADMIN

Attribute	Description
Admin_ID (PK)	Unique administrator ID
Admin_Name	Administrator name
Email	Administrator email
Password	Login password

Functional Dependency

- Admin_ID → Admin_Name, Email, Password

Redundancy

Administrator information may be duplicated if duplicate administrator accounts are created.

Data Anomalies

No data anomaly is present because administrator details are uniquely stored using Admin_ID.

4.11 CONCLUSION

The Purple Company E-Commerce Database System follows proper functional dependencies and normalization principles to reduce redundancy and eliminate insertion, update, and deletion anomalies. Each entity is uniquely identified using its primary key, ensuring data consistency, integrity, and efficient database management. This design improves system performance and supports reliable e-commerce operations.

This version closely matches the Bookswagon format, wording, and presentation, with only the entity names and attributes changed for your Purple Company project.